

Requirement Analysis

Technology Stack (Architecture & Stack)

Date	30 September 2026
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	3 Marks

Technical Architecture:

The application uses the ServiceNow platform to implement record-level security using Access Control Lists (ACLs). Users are assigned specific roles, and access to records is controlled based on the Branch field value. Administrators have full access, while authorized users are allowed to view, create, write, and delete records according to their assigned roles.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interface used by users to access and manage institution/student records.	ServiceNow User Interface
2	Application Logic-1	Logic used to control record read access based on user role and Branch value.	JavaScript
3	Application Logic-2	Logic used to check user roles and provide administrative access.	ServiceNow GlideSystem (gs)
4	Application Logic-3	Logic used to implement record-level access through ACL rules.	ServiceNow ACL Script
5	Database	Stores institution/student details including roll number, name, faculty, branch, email and phone number.	ServiceNow Custom Table
6	Cloud Database	Database service used by the application.	ServiceNow Platform Database

7	File Storage	File storage requirement for the application.	ServiceNow Platform Storage
8	External API-1	External API is not required for this project.	Not Applicable
9	External API-2	External API is not required for this project.	Not Applicable
10	Machine Learning Model	Machine learning is not used in this project.	Not Applicable
11	Infrastructure (Server / Cloud)	Application is developed and deployed on the ServiceNow platform.	ServiceNow Cloud Platform

The project uses a custom u_institution_details table with fields such as Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source framework is used. The project uses the built-in ServiceNow platform and ACL functionality.	ServiceNow Platform
2	Security	Record-level security is implemented using READ, CREATE, WRITE and DELETE ACLs. User roles and Branch conditions are used to control access.	ServiceNow ACL, JavaScript, Roles
3	Scalable Architecture	The application uses the ServiceNow platform architecture, where access control can be applied to records through configurable ACL rules and roles.	ServiceNow Platform

4	Availability	The application is accessed through the ServiceNow platform and its platform-based application environment.	ServiceNow Platform
5	Performance	Access decisions are performed using ACL rules, user roles and record conditions before allowing operations on records.	ServiceNow ACL, JavaScript

The project defines separate ACL permissions for read, create, write and delete operations using roles bb1, bb2, bb3 and bb4.